S:

$i, j := 1, 1;$

$\{a = a' \wedge i = 1 \wedge j = 1\} \to Q'$

parbegin $S_1 \| S_2$ parend

$\}\, S_A$

$\}\, S_B$

S_1:

$\{Inv\} \to Q_1$

while $(i \le n)$ $\to \pi_1$ do

$\{Inv \wedge i \le n\} \to \beta_1$

S_{10}: await $(i = j)$ then

$x, i := a[i], i+1$

ta

$\{Inv\}$

od

$\{Inv \wedge i = n+1\} \to R_1$

S_2:

$\{Inv\} \to Q_2$

while $j \le n$ do

$\{Inv \wedge j \le n\}$

await $i > j$ then

$b[j], j := x, j+1$

ta

$\{Inv\}$

od

$\{Inv \wedge j = n+1\} \to R_2$

FELADAT

$A = (a : \mathbb{Z}^n, b : \mathbb{Z}^n)$

$B = (a' : \mathbb{Z}^n)$

$Q = (a = a')$

$R = (Q \wedge b = a')$

1// $Q \Rightarrow wp(S, R)$

helyett

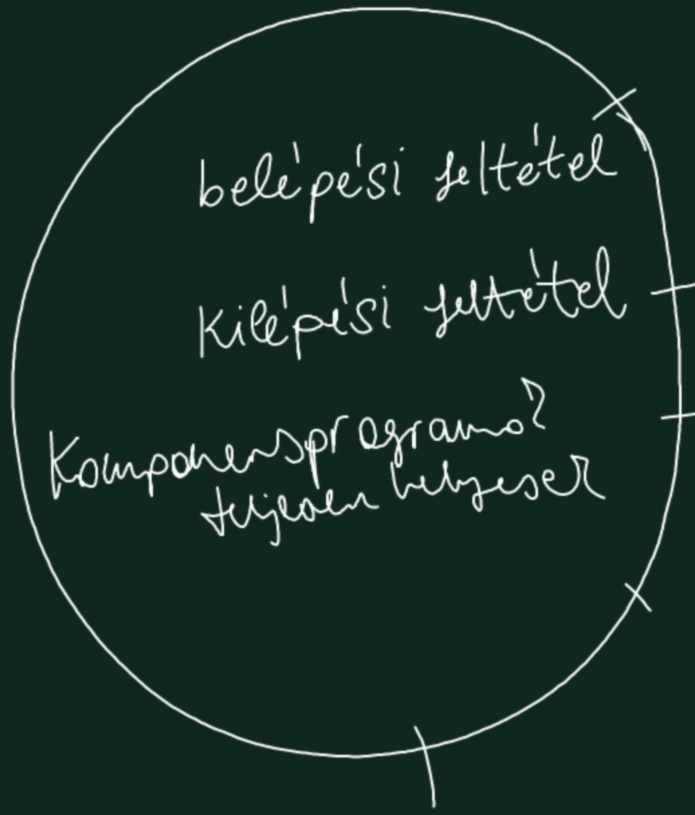
(A) $Q \Rightarrow wp(S_1, Q')$

$Q' \Rightarrow wp(S_2, R)$ → párhuzamos blokk

(A) $wp(i,j := 1,1,\ a=a' \wedge i=1 \wedge j=1) = (a=a' \wedge i=1 \wedge j=1)^{i \leftarrow 1,\, j \leftarrow 1} =$

$= (a=a' \wedge 1=1 \wedge 1=1) = (a=a') = Q$

$Q \Rightarrow Q$ ✓



$$Q \Rightarrow \bigwedge_{i=1}^{n} Q_i$$

$$R_1 \wedge \ldots \wedge R_n \Rightarrow R$$

$$\forall i \in [1..n]: Q_i \Rightarrow \mathcal{G}(S_i, R_i)$$

$$Q_1 \Rightarrow \mathcal{G}(S_1, R_1), \ldots, Q_n \Rightarrow \mathcal{G}(S_n, R_n)$$

teljes helyességi formulák interferenciamentesek

párhuzamos blokk holtpontmentes

$$Q' \Rightarrow Q_1 \wedge Q_2$$

$$\text{Inv} \wedge \text{Inv}$$

$$\text{Inv}$$

$a = a' \checkmark_a$ $\quad$ $a = a' \checkmark_a$

$i = 1 \checkmark_b \checkmark_d$ $\quad$ $0 \leq \underbrace{i-1}_{\underbrace{1-1}_{0}} \leq \underbrace{j}_{1} \leq \underbrace{i}_{1} \leq \underbrace{n+1}_{m,\ \text{ahol } m \geq 1,\ \text{mert } n \geq 0,\ n+1 \geq 1} \checkmark_b$

$j = 1 \checkmark_b \checkmark_c \checkmark_d$

$$\forall k \in [1..\underbrace{j-1}_{\underbrace{1-1}_{0}}) : b[k] = a[k] \checkmark_c$$

igaz

$$\underbrace{i > j}_{\underbrace{1>1}_{\text{hamis}}} \rightarrow x = a[i-1] \checkmark_d$$

igaz

$\text{Inv} \wedge i = n+1 \wedge j = n+1 \Longrightarrow Q \wedge b = a'$

$a = a' \checkmark_a$

$0 \leq i-1 \leq j \leq i \leq n+1$

$\forall k \in [1..j-1] : a[k] = b[k]$

$i > j \rightarrow x = a[i-1]$

$i = n+1$

$j = n+1 \checkmark_b$

$a = a' \checkmark_a$

$b = a' \checkmark_b$

(b) $j = n+1 \rightarrow n = j-1$

$\forall k \in [1..\underbrace{j-1}_{n}] : a[k] = b[k]$

$a' = a = b$

$Q_1 \Rightarrow ef(S_1, R_1)$

$Q_2 \Rightarrow ef(S_2, R_2) \rightarrow$ nem nézzük meg

$Inv \Rightarrow ef(S_1, Inv \wedge i = n+1)$

helyett

$Inv \Rightarrow Inv$ (ez a „P")

$Inv \Rightarrow \Pi_1 \wedge \neg \Pi_1$

$Inv \wedge \neg \Pi_1 \Rightarrow R_1$

$Inv \wedge \Pi_1 \Rightarrow t > 0$ $\rightarrow n+1-i$

$Inv \wedge \Pi_1 \wedge t = t_0 \Rightarrow ef(S_{1,0}, Inv \wedge t < t_0)$ $(t_0 \in \mathbb{Z})$

$Inv \Rightarrow Inv$ ✓

$Inv \Rightarrow i \leq n \vee i > n$ ✓a

ⓐ $i : \mathbb{N}, j : \mathbb{N}$, tehát $i \leq n \vee i > n$

$Inv \wedge \neg \Pi_1 \Rightarrow R_1$

$a = a'$ ✓a

$0 \leq i-1 \leq j \leq i \leq n+1$ ✓b ✓e

$\forall k \in [1..j-1] : a[k] = b[k]$ ✓c

$i > j \rightarrow x = a[i-1]$ ✓d

$i > n$ ✓e

(e) $n < i \leq n+1$, teda $i \in [n+1..n+1]$, teda $i = n+1$

$a = a'$ ✓a

-||- ✓b

-||- ✓c

-||- ✓d

$i = n+1$ ✓e

$Inv \wedge \Pi_1 \Rightarrow t > 0$

$i \leq n$ ✓a $\quad n+1-i > 0$ ✓a

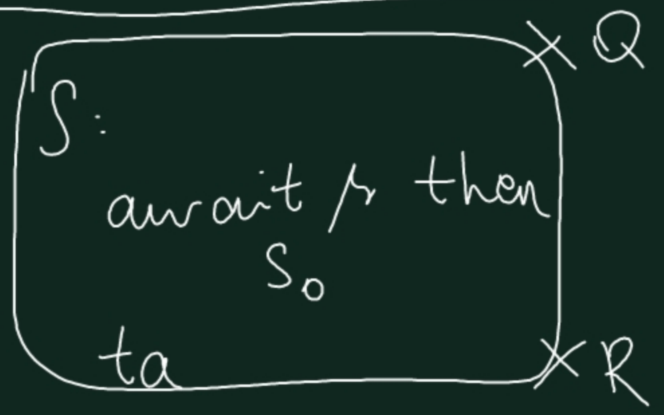
⋮

(a) $i \leq n$, teda

$0 \leq n-i$,

$0 < n+1-i$

Várakozó utasítás levezetési szabálya



$Q \Rightarrow wp(S, R)$

helyett

$Q \Rightarrow \beta \vee \neg\beta$

$Q \wedge \beta \Rightarrow wp(S_0, R)$

S legyen párh. blokkban, ahol vizsgáljuk a holtpontmentességet.

$Inv \wedge T_1 \wedge t = t_0 \Rightarrow i = j \vee i \neq j$ ✓a

(a) $i : \mathbb{N}, j : \mathbb{N}$, tehát $i = j \vee i \neq j$

$Inv \wedge T_1 \wedge t = t_0 \wedge i = j \Rightarrow$

$wp(x, i := a[i], i+1, Inv \wedge t < t_0)$

$wp(x, i := a[i], i+1, Inv \wedge t < t_0) =$

$= (a = a', 0 \leq i \leq j \leq i+1 \leq n+1,$

$\forall k \in [1..j-1] : a[k] = b[k], i+1 > j \Rightarrow$

$a[i] = a[i], n - i < t_0, i \in [1..n])$

$Inv \wedge \Pi_1 \wedge t = t_0 \wedge i = j \Longrightarrow$ eg $(x, i := a[i], i+1, Inv \wedge t < t_0)$

$a = a'$ ✓a

$0 \leq i-1 \leq j \leq i \leq n+1$ ✓b ✓d ✓f

$\forall k \in [1..j-1]: b[k] = a[k]$ ✓c

$i > j \rightarrow x = a[i-1]$

$i \leq n$ ✓b ✓d ✓f

$n+1-i = t_0$ ✓e

$i = j$ ✓b

$a = a'$ ✓a

$0 \leq i \leq j \leq i+1 \leq n+1$ ✓b

$\forall k \in [1..j-1]: b[k] = a[k]$ ✓c

$i+1 > j \rightarrow a[i] = a[i]$ ✓d

$n - i < t_0$ ✓e

$i \in [1..n]$ ✓f

(d) $a[i] = a[i]$ igaz,

$0 \leq i-1$, tehát $1 \leq i$ → (f)

$i \leq n$ → $i \in [1..n]$

(b) $i = j+1$

$0 \leq i-1 \leq j \leq i$

$0 \leq i \leq j \leq i+1$

$i \leq n$

$i+1 \leq n+1$

(e) $n+1-i = t_0$

$\downarrow -1$

$n - i < t_0$